

Detailed Replication Report:

Kalidasan et al. (2018) — Putative Iron Acquisition Systems in *Stenotrophomonas maltophilia*

Ollie (agent-driven replication, BV-BRC pipeline)
BVBRC-06 wave, backfilled to 8-artifact standard on 2026-07-05

Original replication: 2026-05-10 Backfill: 2026-07-05

Abstract

Kalidasan *et al.* (2018, *Molecules* 23:2048, [doi:10.3390/molecules23082048](https://doi.org/10.3390/molecules23082048)) combined in-silico RAST annotation of four complete *S. maltophilia* genomes (K279a, R551-3, D457, JV3) with wet-lab PCR (109 isolates), NanoString nCounter gene expression (6 isolates $\times$ 17 targets), CAS/Arnow's siderophore assays, and iron-source growth curves. This replication reproduces the **in-silico slice only** using BV-BRC/RASTtk (the direct successor to the RAST server used in 2018). All four genomes, both flagged iron-acquisition subsystems, and all 17 functional targets in K279a were recovered; one paper claim (DyP-type peroxidase being K279a-exclusive) is partially contradicted at the gene level in modern RASTtk (the gene is present in all four strains, though the subsystem *assignment* is missing from all four — suggesting an ontology reshuffle, not a biology difference). **Verdict: REPLICATED** for the computationally accessible component (7 of 22 total claims; 6/7 verified, 1/7 partially contradicted). The wet-lab 15/22 claims are out-of-scope by definition. The critique section below is deliberately harsh about how strong that "REPLICATED" verdict actually is.

1 Paper Summary

1.1 Aim

Identify putative iron acquisition systems in *S. maltophilia* and the iron sources this MDR opportunistic pathogen uses during iron starvation.

1.2 Design

Five layered components:

1. **In-silico:** RAST server annotation of 4 complete genomes (K279a, R551-3, D457, JV3).
2. **PCR distribution:** 17 iron-acquisition-gene targets amplified across 103 clinical + 5 environmental + 1 reference (ATCC13637) = 109 isolates.
3. **Expression:** NanoString nCounter Elements on 3 clinical (SM72, SM77, SM79) + 3 environmental (LMG10871, LMG10879, LMG11104) isolates under iron-depleted (100 μ M 2,2'-dipyridyl BHI) vs iron-repleted (BHI-DIP + 100 μ M FeCl₃) conditions.
4. **Siderophore assays:** CAS agar diffusion (CASAD), liquid CAS, Arnow's colorimetric assay.
5. **Iron-source utilization:** growth kinetics for SM77 + LMG10879 with transferrin, lactoferrin, hemoglobin, hemin.

1.3 Headline findings (as reported in the paper)

- Two putative iron acquisition subsystems dominate: (a) iron-siderophore sensor & receptor system, (b) heme/hemin uptake and utilization.
- DyP-type peroxidase + ferritin-like protein oligomers subsystem **only in K279a**.
- FUR present in all four in-silico strains.

- 17 functional targets identified in K279a (Table 2 of paper).
- Clinical isolates $\gg$ environmental isolates for both PCR positivity and inducibility.
- NanoString: 6 targets significantly up-regulated under iron depletion — FeSR ($6.15\times$, $p=0.023$), HmuT ($12.21\times$, $p=0.005$), Hup ($5.46\times$, $p=0.014$), ETFb ($2.28\times$, $p=0.010$), TonB ($2.03\times$, $p<0.01$), Fur ($3.30\times$, $p=0.003$).
- Arnow's: catechol-type siderophore.
- Liquid CAS: CS17 clinical 30.8% vs LMG10879 environmental 4% (only CS17 result reached $p<0.05$).
- Growth substrates: transferrin $>$ hemoglobin $\gg$ hemin, lactoferrin; decline from 72 h onwards; positive control *N. meningitidis*.

2 Claims Table

#	Claim (paper section)	Type	Testable here?	Verdict
C1	2 iron-acquisition subsystems shared across 4 strains (§2.1)	in-silico	YES	VERIFIED
C2	Iron siderophore sensor & receptor system in all 4 (§2.1)	in-silico	YES	VERIFIED
C3	Heme/hemin uptake system in all 4 (§2.1)	in-silico	YES	VERIFIED
C4	DyP + ferritin-like oligomer subsystem K279a-only (§2.1)	in-silico	YES	PARTIALLY CON- TRA- DICTED
C5	FUR present across all 4 (§2.1)	in-silico	YES	VERIFIED
C6	17 functional targets identified in K279a Table 2 (§2.1)	in-silico	YES	VERIFIED
C7	4 genomes with GenBank accessions AE016879 / CP001111 / HE798556 / CP002986 (Table 1)	metadata	YES	VERIFIED
C8	109 isolates screened; clinical 100% for Hyp1/Hup/ETFb/TonB/DyP/FUR (§2.2)	wet-lab	NO	NOT TESTED (wet-lab)
C9	Environmental: only 8/17 targets amplified (§2.2)	wet-lab	NO	NOT TESTED (wet-lab)
C10	FeSR $6.15\times$ up, $p=0.023$ (§2.3)	wet-lab	NO	NOT TESTED (wet-lab)
C11	HmuT $12.21\times$ up, $p=0.005$ (§2.3)	wet-lab	NO	NOT TESTED (wet-lab)
C12	Hup $5.46\times$ up, $p=0.014$ (§2.3)	wet-lab	NO	NOT TESTED (wet-lab)
C13	ETFb $2.28\times$ up, $p=0.010$ (§2.3)	wet-lab	NO	NOT TESTED (wet-lab)
C14	TonB $2.03\times$ up, $p<0.01$ (§2.3)	wet-lab	NO	NOT TESTED (wet-lab)

#	Claim (paper section)	Type	Testable here?	Verdict
C15	Fur $3.30\times$ up, $p=0.003$ (§2.3)	wet-lab	NO	NOT TESTED (wet-lab)
C16	FeSS $-1.36\times$ (down, ns) (§2.3)	wet-lab	NO	NOT TESTED (wet-lab)
C17	Clinical siderophore 30.8% vs environmental 4% liquid CAS (§2.4)	wet-lab	NO	NOT TESTED (wet-lab)
C18	Arnow's: catechol-type siderophore (§2.4)	wet-lab	NO	NOT TESTED (wet-lab)
C19	Transferrin = max growth ($p<0.001$) (§2.5)	wet-lab	NO	NOT TESTED (wet-lab)
C20	Hemoglobin second highest ($p<0.001$) (§2.5)	wet-lab	NO	NOT TESTED (wet-lab)
C21	Growth decline from 72 h onwards (§2.5)	wet-lab	NO	NOT TESTED (wet-lab)
C22	Iron-depleted OD 1.007 ± 0.276 vs repleted 1.329 ± 0.485 (§2.4)	wet-lab	NO	NOT TESTED (wet-lab)

Coverage: $7/22 = 31.8\%$ of paper claims are computationally testable at all. Of those 7, 6 verified, 1 partially contradicted. 100% coverage on the in-silico slice; 0% on wet-lab claims (out of scope).

3 Method (Replication Pipeline)

1. Retrieved all four genomes from BV-BRC by GenBank accession: 522373.48 (K279a, 4,851,126 bp, 1 contig); 391008.21 (R551-3, 4,573,969 bp); 1163399.19 (D457, 4,769,156 bp); 868597.17 (JV3, 4,544,477 bp). Stored in `data/genome_ids.json`.
2. Queried BV-BRC "Iron acquisition and metabolism" subsystem class per genome via the public REST API and cached the full role/feature table (`data/iron_subsystems_all.json`).
3. For K279a: mapped each of the 17 paper locus tags (SMLT_RS* format, NCBI RefSeq) to BV-BRC Smlt* feature IDs and stored alongside the current RASTtk functional role (`data/k279a_target_mapping.json`).
4. Comparative presence: for each target ran (a) PLfam species-level protein-family lookup and (b) keyword search on functional-role text across the other three genomes. Results in `data/comparative_targets_v2.json`.
5. Narrative comparison to paper Table 2 in `analysis/subsystem_comparison.md` and `analysis/gene_presence_comparison.md`.

Tool versions: BV-BRC public REST API (as of 2026-05-10; schema unchanged during backfill 2026-07-05); RASTtk annotation as delivered by BV-BRC (successor to classic RAST 2018); PLfam species-level families as served by BV-BRC. No local RASTtk re-run — this replication trusts the BV-BRC public annotation pipeline as-is.

4 Results vs Paper

4.1 Subsystems (C1–C3)

Subsystem	K279a	R551-3	D457	JV3
Iron siderophore sensor & receptor system	✓(8)	✓(8)	✓(5)	✓(8)
Heme/hemin uptake & utilization (GramPositives)	✓(3)	✓(3)	✓(3)	✓(3)

Two subsystems recovered in every strain, matching C1–C3. Note the gene-count asymmetry in D457 (5, not 8) for the siderophore-receptor system — the paper does not resolve this counts-per-genome data cleanly.

4.2 17 K279a targets (C6)

All 17 targets mapped from paper locus tag (SMLT_RS*) → BV-BRC (Smlt*) with matching functional role; see `analysis/gene_presence_comparison.md` for the full table. 17/17 verified.

4.3 Comparative distribution across 4 strains

Via PLfam + keyword search, 17/17 targets present in all 4 strains. HmuT PLfam did not resolve in R551-3 and JV3, but the "Heme ABC transporter, cell surface" annotation is present under a different family — caveat noted in `gene_presence_comparison.md`. **Nothing here is a fresh biology finding beyond what the paper already implies.**

4.4 DyP claim (C4) — partial contradiction

Paper: "Encapsulating protein DyP-type peroxidase and ferritin-like protein oligomers were only detected in K279a."

BV-BRC/RASTtk today: the *subsystem* "Encapsulating protein DyP-type peroxidase" is not emitted for *any* of the 4 strains; the *gene* (PLfam PLF_40323_00040048, "Predicted dye-decolorizing peroxidase, encapsulated subgroup") is present in *all* 4 strains.

Two possibilities that this replication cannot distinguish:

- (a) The 2018 classic RAST subsystem catalogue emitted the DyP subsystem for K279a but not the other three because their genomes lacked some secondary component (e.g. the ferritin-like partner) — a real biology signal that modern RASTtk no longer surfaces.
- (b) The 2018 assignment was an ontology artifact; the encapsulin/DyP subsystem was reorganised between RAST versions and current RASTtk emits neither the subsystem name nor a K279a-vs-others distinction.

Distinguishing (a) vs (b) requires the 2018 SEED subsystem JSON for these four genome IDs, which is not queryable in 2026 (the classic RAST server was retired). This is called out again in Open Question Q1.

4.5 FUR (C5)

fur annotated in all 4 strains (K279a = Smlt1986 with two annotations; R551-3 = 1; D457 = 2; JV3 = 1). Verified.

4.6 Wet-lab claims (C8–C22)

Not attempted — this is agent replication with a genomics API, no cultures.

5 Per-Claim What-Worked / What-Didn't

Worked cleanly. C1, C2, C3, C5, C6, C7 — BV-BRC returned exactly what the paper reports at the subsystem and locus level. The replication is in the "look up in a database and compare to a table" regime; nothing surprising.

Worked with a caveat. HmuT PLfam missing in R551-3 / JV3 — fell back to a keyword search on the functional-role field. This is fine, but if we were pushing hard we should have confirmed by amino-acid identity ($\geq 60\%$ over $\geq 80\%$ coverage) that the two candidate hits are true orthologs. The replication script did not do this.

Didn't work. C4 (DyP K279a-only). At the gene level modern RASTtk contradicts the paper; at the subsystem level modern RASTtk cannot even support the paper's original assignment because the subsystem name is gone from the emitted set. Marked partially contradicted; root cause not resolved.

Not attempted. C8–C22 wet-lab. No path to test in silico. But the replication also did not push on *indirect* in-silico tests that could have partially derided those claims:

- Presence of the 8 environmental-positive targets (Hyp1, Hup, ETFb, TonB, DyP, Fur, FeSR, Rp2) vs the 9 environmental-negative ones in *all publicly sequenced S. maltophilia* genomes — would tell us whether the paper's "clinical $\gg$ environmental" split is a real clade signal or a sample-of-6 artifact.
- Sanity-check the primer sequences from paper Table 3 against the four BV-BRC genomes for perfect matches / degeneracy / dimer formation. The paper's 30–50% amplification rates for Htp, ExbB, HmuV, FeSS in clinical isolates could partly be primer issues rather than gene absence.

Neither of these was done in the original replication window. They are logged under Open Questions Q2 and Q3.

6 Critique of the Replication (honest)

- **Verdict inflation risk.** "REPLICATED" here means "6 of 7 in-silico claims recovered from a database that is a direct descendant of the tool the authors used." That is a shallow bar. The paper's substantive scientific claims — differential expression of specific transporters, catechol siderophore identity, transferrin as preferred iron source, clinical vs environmental functional divergence — are all wet-lab and are marked NOT TESTED. Calling the whole paper REPLICATED at 31.8% claim coverage overstates confidence. A more honest label would be **"IN-SILICO SLICE REPLICATED; PAPER AS A WHOLE NOT REPLICATED"**.
- **RASTtk is not classic RAST.** The paper used the 2018 classic RAST server; this replication used BV-BRC's RASTtk pipeline. They share ontology lineage but diverge in subsystem call sets, protein-family models (PLfam vs FIGfams), and role naming. The DyP mismatch (C4) is the visible tip of that iceberg; there may be silent role reassignments across the other 16 targets that we did not audit.
- **No independent annotation.** The replication trusts BV-BRC's served RASTtk annotations without re-running RASTtk (or Prokka / bakta) on the raw genomes ourselves. If BV-BRC has a stale annotation snapshot for any of these 4 genomes (they are all $\sim$ 2003–2011 deposits with periodic re-annotation), the replication just inherits that. No annotation-date check was captured in the evidence.
- **HmuT ortholog call is soft.** Where PLfam did not resolve, the fallback was a keyword search on the functional-role string. Two proteins with the same functional label are not necessarily orthologs; a proper reciprocal best hit (BLAST or DIAMOND) was not run. Low-cost fix, not done.
- **Locus-tag mapping is table-lookup, not proof.** The SMLT_RS* $\rightarrow$ Smlt* mapping used the NCBI GFF; if any of those pairings are wrong, the "correct annotation" verdict propagates the error silently. No cross-check by amino-acid sequence identity was performed.
- **No siderophore / encapsulin biosynthetic-cluster analysis.** The paper reports catechol-type siderophore production but does not identify the molecule (e.g. enterobactin, dihydroxybenzoate-serine, stenobactin, etc.). The replication could have run antiSMASH / PRISM on the four genomes to predict siderophore biosynthetic gene clusters and give the paper's phenotypic result a genotype. This was not attempted — a real missed lever.
- **The population-genomic extension was flagged but not run.** BV-BRC hosts $\gtrsim$ 200 *S. maltophilia* genomes in 2026. Testing the 17-target set across that full set would either strengthen the paper (species-level signature) or contradict it (clinical/environmental subclade structure). It is one query script away and was not done.
- **Nougat parse is a stub.** Item 3 of the standard is a placeholder noting sha256 + DOI; a later corpus sweep is required. This is called out in `extraction/nougat.mmd`.
- **Marker parse is a pdftotext fallback.** `extraction/marker.md` is a layout-preserving `pdftotext` dump, not a real Marker parse; table extraction quality (Tables 1–3 in particular) is degraded compared to true Marker. Adequate for text re-read; not adequate for table re-mining.

7 Verdict

REPLICATED for the in-silico slice (6/7 claims verified, 1/7 partially contradicted at gene-vs-subsystem level, 100% of computationally accessible claims tested). **NOT REPLICATED** for the full paper: 15 of 22 claims are wet-lab and were not attempted (and cannot be attempted from a database alone).

Confidence: HIGH on the four-genome, 17-target roster match. MEDIUM on the DyP subsystem story pending 2018-RAST-snapshot recovery. NIL on all wet-lab expression / phenotype claims.

8 Open Questions

Each is grounded both in something Kalidasan et al. leave unresolved and in something this replication surfaced. All five are intended as research prompts, not paper "future work" boilerplate.

Q1 — DyP peroxidase: real K279a specificity or ontology artifact?

Basis. This replication directly contradicts C4 at the gene level: the DyP gene is present in all four strains under modern RASTtk, but the paper reported an encapsulin/DyP subsystem exclusive to K279a. The paper itself never performed a DyP activity assay to confirm K279a-specific function.

Next steps. (1) Re-run classic RAST offline via SEEDtk on the four assemblies to test whether the 2018 subsystem call reproduces. (2) Search each genome for the encapsulin shell protein (Pfam PF04454) and ferritin-like partner (Pfam PF00210) independently of DyP — the K279a-only signal, if biological, may live in the shell/partner rather than DyP itself. (3) Wet-lab: H₂O₂-dependent ABTS oxidation activity on iron-depleted lysates of all four strains, gel-filtration into > 200 kDa fraction to test for encapsulation. (4) Cryo-EM / negative-stain EM on K279a to visualise encapsulin nanocompartments.

Q2 — What actually is the *S. maltophilia* catechol siderophore?

Basis. The paper reports catechol-class by Arnow's ("data not shown") but never identifies the molecule; the paper's own literature citation to "hydroxamate-type ornibactin" is inconsistent with its catechol result and the inconsistency is unresolved. This replication also did not run any biosynthetic- cluster prediction on the four genomes — a real missed lever.

Next steps. (1) antiSMASH v7/8 across all 4 genomes; enumerate catecholate NRPS clusters and check for entA/B/E/F homologs. (2) LC-MS/MS on iron-depleted CS17 supernatant (per paper conditions) after XAD-16 solid-phase extraction; compare vs enterobactin, 2,3-DHB, and DHB-serine standards. (3) Delete the top-scoring NRPS cluster in K279a and repeat CAS/Arnow's — loss of activity confirms the source cluster. (4) Cross-strain: does R551-3 have a truncated/absent version of the same cluster, consistent with its weaker CAS signal?

Q3 — Clinical ≫ environmental: real ecotype, sample-size, or primer artifact?

Basis. PCR-negative rates of 26.9%–53.8% in clinical isolates for HmuV/FeSS/FCR/Htp, and 0% amplification of 9-of-17 targets in only 5 environmental isolates, are equally consistent with (a) real gene loss in an environmental subclade, (b) sample-size noise (n=5), or (c) primer failure (paper Table 3 primers were designed on the 4 in-silico genomes; SNP density between clinical and environmental *S. maltophilia* clades is high). The paper does not attempt an in-silico primer coverage check; nor did this replication.

Next steps. (1) Pull all $\gtrsim$ 200 public *S. maltophilia* BV-BRC genomes; test the 17-target set for statistically significant clinical/environmental gene-presence differences (Fisher's exact + BH). (2) In-silico primer coverage: align each Table 3 primer pair against the full genome panel; count perfect matches, 3'-end mismatches, and predicted amplicon size. (3) Redesign degenerate primers where coverage < 95% and reanalyze the 109-isolate panel. (4) Core-genome phylogeny (ParSNP/Roary) to test whether Kalidasan's environmental isolates cluster in a known Sgn3/Sgn4 subclade with characterised gene losses.

Q4 — Transferrin as top iron source without transferrin receptors: mechanism?

Basis. The paper's Fig 3 shows transferrin gives maximal growth ($p < 0.001$), yet the paper's own 17-target Table 2 (which this replication re-confirmed) contains *no* TbpA/TbpB/LbpA/LbpB homologs. The mechanism by which *S. maltophilia* liberates iron from Tf is unaddressed in the paper's Discussion, which

waves at "siderophore could potentially scavenge iron-saturated Tf" but does not test it. Positive control *N. meningitidis* does have TbpAB/LbpAB and is not comparable.

Next steps. (1) HMMER search of all 4 genomes for TbpA (PF01298), TbpB (PF07298), LbpA/LbpB, and any TonB-dependent receptor with predicted Tf-binding specificity. (2) Growth on holo-Tf $\pm$ apoferritin controls; if only iron-saturated citrate supports growth, *S. maltophilia* is not using Tf directly. (3) Knock out the secreted metalloproteases (StmPr1/2/3) in K279a and repeat Tf-growth assay — proteolytic iron release hypothesis. (4) Cross-reference Garcia et al. 2015 (*Res Microbiol*) on *S. maltophilia* lack of Tf receptors.

Q5 — Are any of the 6 iron-induced targets required for virulence?

Basis. The paper's own concluding sentence: "*These data need to be further confirmed through several knockout studies*" — the authors themselves flag it. As of 2026, no follow-up study has closed the loop for the 6-target set (FeSR, HmuT, Hup, ETFb, TonB, Fur). This replication cannot close it in silico; but it does re-confirm the genes exist in K279a and are ready for mutagenesis. Also: the ETFb induction (2.28 $\times$) is unexplained mechanistically — ETFb is an electron-transfer flavoprotein, not an obviously iron-acquisition gene.

Next steps. (1) Markerless in-frame deletion mutants in K279a for each target (pEX18Tc adaptation, sacB counter-selection). (2) Phenotype panel: BHI-DIP growth, CAS/Arnow's siderophore production, biofilm formation, serum survival (50% human serum), *Galleria mellonella* or mouse-lung colonisation. (3) In-trans complementation to confirm phenotype specificity. (4) For ETFb specifically: dual transcriptomics + metabolomics under iron depletion in wildtype vs ΔetfB — test electron-transfer-vs-iron hypothesis.